

Advanced Computer Networks LAB

Spring Semester 2026

Assignment 4

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Roll No – 22CS02002

1. Packet Identification and Addressing –

No.	Time	Source	Destination	Protocol	Length	Info
25	141.770	172.25.1.106	180.149.59.146	ICMP	98	Echo (ping) request id=0x0002, seq=4/1024, ttl=64 (reply in 26)
26	141.803	180.149.59.146	172.25.1.106	ICMP	98	Echo (ping) reply id=0x0002, seq=4/1024, ttl=56 (request in 25)
27	142.773	172.25.1.106	180.149.59.146	ICMP	98	Echo (ping) request id=0x0002, seq=5/1280, ttl=64 (reply in 28)
28	142.805	180.149.59.146	172.25.1.106	ICMP	98	Echo (ping) reply id=0x0002, seq=5/1280, ttl=56 (request in 27)
29	143.493	Microsoft_a7	Microsoft_id	ARP	42	Who has 172.25.1.106? Tell 172.25.0.1
30	143.493	Microsoft_id	Microsoft_a7	ARP	42	172.25.1.106 is at 00:15:5d:1d:27:b9
31	151.506	172.25.0.1	172.25.15.2	NBNS	92	Name query NB DESKTOP-RHVH0B7<1c>
32	152.326	172.25.0.1	172.25.15.2	NBNS	92	Name query NB DESKTOP-RHVH0B7<1c>
33	153.147	172.25.0.1	172.25.15.2	NBNS	92	Name query NB DESKTOP-RHVH0B7<1c>
34	156.755	172.25.1.106	91.189.91.1	NTP	90	NTP Version 4, client
35	159.787	172.25.1.106	184.18.34.2	ICMP	98	Echo (ping) request id=0x0003, seq=1/256, ttl=64 (reply in 36)
36	159.835	184.18.34.2	172.25.1.106	ICMP	98	Echo (ping) reply id=0x0003, seq=1/256, ttl=52 (request in 35)

Frame 27: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0
Ethernet II, Src: Microsoft_a7:6b:38 (00:15:5d:1d:27:b9), Dst: Microsoft_a7:6b:38 (00:15:5d:a7:6b:38)
Source: Microsoft_id:27:b9 (00:15:5d:1d:27:b9)
Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 172.25.1.106, Dst: 180.149.59.146
0100 ... = Version: 4
... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 84
Identification: 0x41b4 (16820)
Flags: 0x40, Don't fragment
... 0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0x5b4a (validation disabled)
[Header checksum status: Unverified]
Source Address: 172.25.1.106
Destination Address: 180.149.59.146
Internet Control Message Protocol

0000 00 15 5d a7 6b 38 00 15 5d 1d 27 b9 08 00 45 00 ...] k8 ...] ... E
0010 00 54 41 b4 00 40 91 00 4a 00 00 00 04 95 ... TA @ @ : ...
0020 3b 92 08 00 87 91 00 02 00 05 b5 05 db 69 00 00 ; 1 ..

eth0: <live capture in progress> Packets: 46 - Displayed: 46 (100.0%) Profile: Default

2. Traffic Clustering and Analysis –

Ans – 1)

(a) Local Network (Same Subnet) → 82%

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	162.159.198.2	10.10.92.57	TLSv1.2	311	Application Data
3	0.007253	162.159.198.2	10.10.92.57	TLSv1.2	336	Application Data
4	0.014087	162.159.198.2	10.10.92.57	TLSv1.2	311	Application Data
5	0.022737	162.159.198.2	10.10.92.57	TLSv1.2	311	Application Data
6	0.030233	162.159.198.2	10.10.92.57	TLSv1.2	327	Application Data
7	0.037924	180.149.52.98	10.10.92.222	TCP	66	443 → 51676 [ACK] Seq=1 Ack=1 Win=61 Len=0 TSval=4168258962 TSecr=3228316372
8	0.037924	180.149.52.98	10.10.92.222	TLSv1.2	90	Application Data
9	0.037924	180.149.52.98	10.10.92.222	TCP	66	443 → 51676 [FIN, ACK] Seq=25 Ack=2 Win=61 Len=0 TSval=4168258962 TSecr=3228316373
10	0.039113	162.159.198.2	10.10.92.57	TLSv1.2	311	Application Data
11	0.047633	162.159.198.2	10.10.92.57	TLSv1.2	327	Application Data
12	0.052528	142.251.43.174	10.10.92.50	TLSv1.2	211	Application Data
13	0.052528	162.159.198.2	10.10.92.57	TLSv1.2	311	Application Data
14	0.057235	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
15	0.057235	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
16	0.061482	162.159.198.2	10.10.92.57	TLSv1.2	311	Application Data
17	0.068437	162.159.198.2	10.10.92.57	TLSv1.2	327	Application Data
18	0.076254	162.159.198.2	10.10.92.57	TLSv1.2	311	Application Data

Frame 1: 311 bytes on wire (2488 bits), 311 bytes captured (2488 bits) on interface \Device\NPF_{C010246A-3BF0-4D39-AAE3-A940C94EA532}, id 0
Ethernet II, Src: Hewlett-Packard_ba:4a:01 (d8:94:03:ba:4a:01), Dst: Intel_08:c8:48 (10:91:d1:08:c8:48)
Internet Protocol Version 4, Src: 162.159.198.2, Dst: 10.10.92.57
Transmission Control Protocol, Src Port: 443, Dst Port: 6087, Seq: 1, Ack: 1, Len: 245
Transport Layer Security

0000 10 91 d1 08 c
0010 01 29 84 fa 4
0020 5c 39 01 bb 1
0030 96 00 06 53 e
0040 ae 62 17 03 e
0050 9a 55 bc bd 4
0060 b4 09 3a dc 4
0070 74 7e a8 2e e
0080 1c 4e 34 16 c
0090 b3 e7 08 2d e
00a0 41 a6 61 ce e
00b0 d7 fe 44 26 e
00c0 6f 53 02 bd t

wireshark_EthernetOYDBN3.pcapng Packets: 39198 - Displayed: 32154 (82.0%) - Dropped: 0 (0.0%) Profile: Default

Ans – 2) Router MAC = d8:94:03:ba:4a:01

No.	Time	Source	Destination	Protocol	Length	Info
558	2.515167	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=77207 Ack=55093 Win=38400 Len=0 TSval=340171905 TSecr=1226847
551	2.515167	162.159.198.2	10.10.92.57	TLSv1.2	177	Application Data
552	2.525259	18.164.217.72	10.10.93.54	TCP	60	80 → 51500 [FIN, ACK] Seq=1 Ack=1 Win=499 Len=0
553	2.526426	10.10.93.234	34.8.7.18	TCP	55	52053 → 443 [ACK] Seq=1 Ack=1 Win=1021 Len=1
554	2.527177	172.66.0.227	10.10.93.96	TLSv1.2	105	Application Data
555	2.529396	162.159.198.2	10.10.92.57	TLSv1.2	1057	Application Data
556	2.529396	162.159.198.2	10.10.92.57	TLSv1.2	258	Application Data
557	2.532248	44.222.27.214	10.10.93.234	TCP	66	443 → 62383 [ACK] Seq=1 Ack=2 Win=187 Len=0 SLE=1 SRE=2
558	2.537534	HewlettPacka_ba:4a:01	2a:d9:ef:38:d5:61	ARP	60	10.10.92.1 is at d8:94:03:ba:4a:01
559	2.537534	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=78501 Ack=55534 Win=38400 Len=0 TSval=340171927 TSecr=1226865
560	2.539573	162.159.198.2	10.10.92.57	TLSv1.2	344	Application Data
561	2.548254	162.159.198.2	10.10.92.57	TLSv1.2	170	Application Data
562	2.555972	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=78883 Ack=55828 Win=38400 Len=0 TSval=340171946 TSecr=1226882
563	2.558624	142.250.182.78	10.10.92.110	TLSv1.2	120	Application Data
564	2.558624	142.250.182.78	10.10.92.110	TLSv1.2	132	Application Data
565	2.570461	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=78883 Ack=56122 Win=38400 Len=0 TSval=340171960 TSecr=1226898
566	2.570461	34.8.7.18	10.10.93.234	TCP	66	443 → 52053 [ACK] Seq=1 Ack=2 Win=1042 Len=0 SLE=1 SRE=2

(b) 0.9% of packets going to other local machines

No.	Time	Source	Destination	Protocol	Length	Info
14	0.057235	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
15	0.057235	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
101	0.588412	10.10.93.104	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
105	0.624884	10.10.93.104	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _spotify-connect_tcp.local, "QM" question
222	1.011151	10.10.92.181	10.10.93.255	UDP	90	64740 → 59200 Len=48
248	1.069210	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
249	1.069210	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
467	2.080365	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
468	2.080365	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
530	2.402464	10.10.92.42	224.0.0.251	MDNS	137	Standard query 0x0000 PTR _companion-link_tcp.local, "QM" question PTR _rdlink_tcp.l...
540	2.443392	10.10.93.168	224.0.0.251	MDNS	486	Standard query response 0x0000 PTR, cache flush Android_UL838LD4.local AAAA, cache flu...
576	2.653918	10.10.93.104	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
577	2.657748	10.10.93.104	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _spotify-connect_tcp.local, "QM" question
623	3.002309	10.10.92.181	10.10.93.255	UDP	90	64740 → 59200 Len=48
647	3.108466	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
648	3.108466	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
1356	4.104067	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263

> Frame 249: 305 bytes on wire (2440 bits), 305 bytes captured (2440 bits) on interface \Device\NPF_{C010246A-3BF0-4D39-AAE3-A940C94EA532}, id 0

▼ Ethernet II, Src: ASUSTekCOMPU_85:f3:9e (58:11:22:85:f3:9e), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Destination: Broadcast (ff:ff:ff:ff:ff:ff)

> Source: ASUSTekCOMPU_85:f3:9e (58:11:22:85:f3:9e)

Type: IPv4 (0x0800)

[Stream index: 4]

> Internet Protocol Version 4, Src: 10.10.92.139, Dst: 10.10.93.255

> User Datagram Protocol, Src Port: 54915, Dst Port: 54915

> Data (263 bytes)

0000 ff ff ff ff f
0010 01 23 b2 31 e
0020 5d ff d6 83 c
0030 70 74 6f 70 e
0040 00 00 90 ba c
0050 00 00 00 00 e
0060 00 00 00 00 e
0070 00 00 7c 6a e
0080 00 00 d9 b7 4
0090 00 00 20 b5 4
00a0 00 00 40 b4 4
00b0 32 32 36 63 3
00c0 2d 39 66 61 e

wireshark_EthernetOYDBN3.pcapng Packets: 39198 · Displayed: 347 (0.9%) · Dropped: 0 (0.0%) Profile: Default

Ans – 3) My MAC = b4:45:06:7a:81:a7

(a) 5.1% packets where source or destination is my MAC address

No.	Time	Source	Destination	Protocol	Length	Info
353	1.449569	142.251.221.106	10.10.93.234	TLSv1.2	178	Application Data
413	1.696610	192.168.2.190	10.10.93.234	TCP	62	80 → 59037 [SYN, ACK] Seq=0 Ack=1 Win=63980 Len=0 MSS=9140 WS=128
414	1.696678	10.10.93.234	192.168.2.190	TCP	54	59037 → 80 [ACK] Seq=1 Ack=1 Win=1026 Len=0
494	2.244937	10.10.93.234	23.218.230.53	TCP	54	49988 → 443 [FIN, ACK] Seq=1 Ack=1 Win=1026 Len=0
497	2.260722	10.10.93.234	44.222.27.214	TCP	55	62383 → 443 [ACK] Seq=1 Ack=1 Win=1026 Len=1
518	2.348613	23.218.230.53	10.10.93.234	TLSv1.2	85	Encrypted Alert
519	2.348613	23.218.230.53	10.10.93.234	TCP	60	443 → 49988 [FIN, ACK] Seq=32 Ack=2 Win=501 Len=0
520	2.348674	10.10.93.234	23.218.230.53	TCP	54	49988 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
523	2.371919	172.217.24.202	10.10.93.234	TLSv1.2	178	Application Data
525	2.385630	10.10.93.234	44.222.27.214	TCP	55	64568 → 443 [ACK] Seq=1 Ack=1 Win=1025 Len=1
526	2.385769	10.10.93.234	172.217.24.202	TLSv1.2	89	Application Data
527	2.385791	10.10.93.234	172.217.24.202	TLSv1.2	89	Application Data
534	2.412997	172.217.24.202	10.10.93.234	TCP	60	443 → 55120 [ACK] Seq=125 Ack=36 Win=1109 Len=0
535	2.412997	172.217.24.202	10.10.93.234	TCP	60	443 → 55120 [ACK] Seq=125 Ack=71 Win=1109 Len=0
553	2.526426	10.10.93.234	34.8.7.18	TCP	55	52053 → 443 [ACK] Seq=1 Ack=1 Win=1021 Len=1
557	2.532240	44.222.27.214	10.10.93.234	TCP	66	443 → 62383 [ACK] Seq=1 Ack=2 Win=187 Len=0 SLE=1 SRE=2
566	2.570461	34.8.7.18	10.10.93.234	TCP	66	443 → 52053 [ACK] Seq=1 Ack=2 Win=1042 Len=0 SLE=1 SRE=2

> Frame 523: 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface \Device\NPF_{C010246A-3BF0-4D39-AAE3-A940C94EA532}, id 0

▼ Ethernet II, Src: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01), Dst: Dell_7a:81:a7 (b4:45:06:7a:81:a7)

> Destination: Dell_7a:81:a7 (b4:45:06:7a:81:a7)

> Source: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01)

Type: IPv4 (0x0800)

[Stream index: 17]

> Internet Protocol Version 4, Src: 172.217.24.202, Dst: 10.10.93.234

> Transmission Control Protocol, Src Port: 443, Dst Port: 55120, Seq: 1, Ack: 1, Len: 124

> Transport Layer Security

0000 b4 45 06 7a 81
0010 00 a4 e5 28 00
0020 5d ea 01 bb d7
0030 04 55 2b 3e 00
0040 14 74 0f 5d fb
0050 ef 98 46 df db
0060 60 0e 94 65 38
0070 ff 43 25 0a 46
0080 76 6d fb 41 dc
0090 21 7c 09 d6 57
00a0 48 c3 2e f5 1a
00b0 61 c4

Source Hardware Address (eth.src), 6 bytes Packets: 39198 · Displayed: 1994 (5.1%) · Dropped: 0 (0.0%) Profile: Default

(b) 2.8% packets coming from internet to my machine

No.	Time	Source	Destination	Protocol	Length	Info
353	1.449569	142.251.221.106	10.10.93.234	TLSv1.2	178	Application Data
413	1.696610	192.168.2.190	10.10.93.234	TCP	62	80 → 59037 [SYN, ACK] Seq=0 Ack=1 Win=63980 Len=0 MSS=9140 WS=128
518	2.348613	23.218.230.53	10.10.93.234	TLSv1.2	85	Encrypted Alert
519	2.348613	23.218.230.53	10.10.93.234	TCP	60	443 → 49988 [FIN, ACK] Seq=32 Ack=2 Win=501 Len=0
523	2.371919	172.217.24.202	10.10.93.234	TLSv1.2	178	Application Data
534	2.412997	172.217.24.202	10.10.93.234	TCP	60	443 → 55120 [ACK] Seq=125 Ack=36 Win=1109 Len=0
535	2.412997	172.217.24.202	10.10.93.234	TCP	60	443 → 55120 [ACK] Seq=125 Ack=71 Win=1109 Len=0
557	2.532240	44.222.27.214	10.10.93.234	TCP	66	443 → 62383 [ACK] Seq=1 Ack=2 Win=187 Len=0 SLE=1 SRE=2
566	2.570461	34.8.7.18	10.10.93.234	TCP	66	443 → 52053 [ACK] Seq=1 Ack=2 Win=1042 Len=0 SLE=1 SRE=2
579	2.658990	44.222.27.214	10.10.93.234	TCP	66	443 → 64568 [ACK] Seq=1 Ack=2 Win=268 Len=0 SLE=1 SRE=2
604	2.858756	104.18.32.47	10.10.93.234	TCP	60	443 → 61174 [ACK] Seq=1 Ack=1401 Win=19 Len=0
605	2.858756	104.18.32.47	10.10.93.234	TCP	60	443 → 61174 [ACK] Seq=1 Ack=6573 Win=19 Len=0
606	2.858756	104.18.32.47	10.10.93.234	TCP	60	443 → 61174 [ACK] Seq=1 Ack=6874 Win=19 Len=0
628	3.032518	104.18.32.47	10.10.93.234	TLSv1.2	512	Application Data
629	3.032518	104.18.32.47	10.10.93.234	TLSv1.2	101	Application Data
630	3.032518	104.18.32.47	10.10.93.234	TLSv1.2	85	Application Data
642	3.080535	104.18.32.47	10.10.93.234	TCP	60	443 → 61174 [ACK] Seq=537 Ack=6909 Win=19 Len=0

> Frame 523: 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface \Device\NPF_{C010246A-3BF0-4D39-AAE3-A940C94EA532}, id 0
 > Ethernet II, Src: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01), Dst: Dell_7a:81:a7 (b4:45:06:7a:81:a7)
 > Destination: Dell_7a:81:a7 (b4:45:06:7a:81:a7)
 > Source: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01)
 > Type: IPv4 (0x0800)
 > [Stream index: 17]
 > Internet Protocol Version 4, Src: 172.217.24.202, Dst: 10.10.93.234
 > Transmission Control Protocol, Src Port: 443, Dst Port: 55120, Seq: 1, Ack: 1, Len: 124
 > Transport Layer Security

Source Hardware Address (eth.src), 6 bytes | Packets: 39198 · Displayed: 1114 (2.8%) · Dropped: 0 (0.0%) | Profile: Default

3. Protocol-Based Analysis (TCP/UDP) –**Ans – 1) Filters****(a) 97.6% packets using TCP**

No.	Time	Source	Destination	Protocol	Length	Info
514	2.329276	142.250.182.78	10.10.92.110	TCP	60	443 → 53730 [ACK] Seq=283 Ack=3185 Win=8352 Len=0
515	2.333862	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=75055 Ack=51372 Win=38400 Len=0 TSval=340171723 TSecr=1226660
517	2.345579	162.159.198.2	10.10.92.57	TLSv1.2	344	Application Data
518	2.348613	23.218.230.53	10.10.93.234	TLSv1.2	85	Encrypted Alert
519	2.348613	23.218.230.53	10.10.93.234	TCP	60	443 → 49988 [FIN, ACK] Seq=32 Ack=2 Win=501 Len=0
520	2.348674	10.10.93.234	23.218.230.53	TCP	54	49988 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
521	2.353445	162.159.198.2	10.10.92.57	TLSv1.2	180	Application Data
522	2.368395	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=75447 Ack=51960 Win=38400 Len=0 TSval=3401717758 TSecr=1226693
523	2.371919	172.217.24.202	10.10.93.234	TLSv1.2	178	Application Data
524	2.378626	162.159.198.2	10.10.92.57	TLSv1.2	344	Application Data
525	2.385630	10.10.93.234	44.222.27.214	TCP	55	64568 → 443 [ACK] Seq=1 Ack=1 Win=1025 Len=1
526	2.385769	10.10.93.234	172.217.24.202	TLSv1.2	89	Application Data
527	2.385791	10.10.93.234	172.217.24.202	TLSv1.2	89	Application Data
528	2.390502	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=75725 Ack=52401 Win=38400 Len=0 TSval=3401717780 TSecr=1226712
529	2.393824	162.159.198.2	10.10.92.57	TLSv1.2	188	Application Data
532	2.406400	162.159.198.2	10.10.92.57	TCP	66	443 → 6087 [ACK] Seq=75847 Ack=52695 Win=38400 Len=0 TSval=3401717795 TSecr=1226729
533	2.409693	162.159.198.2	10.10.92.57	TLSv1.2	367	Application Data

> Frame 523: 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface \Device\NPF_{C010246A-3BF0-4D39-AAE3-A940C94EA532}, id 0
 > Ethernet II, Src: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01), Dst: Dell_7a:81:a7 (b4:45:06:7a:81:a7)
 > Destination: Dell_7a:81:a7 (b4:45:06:7a:81:a7)
 > Source: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01)
 > Type: IPv4 (0x0800)
 > [Stream index: 17]
 > Internet Protocol Version 4, Src: 172.217.24.202, Dst: 10.10.93.234
 > Transmission Control Protocol, Src Port: 443, Dst Port: 55120, Seq: 1, Ack: 1, Len: 124
 > Transport Layer Security

Transmission Control Protocol: Protocol | Packets: 39198 · Displayed: 38274 (97.6%) · Dropped: 0 (0.0%) | Profile: Default

(b) 1.5% packet using UDP

No.	Time	Source	Destination	Protocol	Length	Info
106	0.624884	fe80::6837:97ff:fe...	ff02::fb	MDNS	107	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
222	1.011151	10.10.92.181	10.10.93.255	UDP	90	64740 → 59200 Len=48
248	1.069210	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
249	1.069210	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
299	1.223477	10.10.0.26	10.10.92.168	DNS	398	Standard query response 0x6dc1 A play.googleapis.com A 216.239.36.223 A 216.239.38.223...
467	2.080365	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
468	2.080365	10.10.92.139	10.10.93.255	UDP	305	54915 → 54915 Len=263
505	2.301556	10.10.0.26	10.10.92.110	DNS	339	Standard query response 0x44eb A chat.google.com A 142.250.205.110 NS ns1.google.com N...
508	2.311084	10.10.0.26	10.10.92.110	DNS	339	Standard query response 0x10d0 A chat.google.com A 142.250.205.110 NS ns2.google.com N...
530	2.402464	10.10.92.42	224.0.0.251	MDNS	137	Standard query 0x0000 PTR _companion-link._tcp.local, "QM" question PTR _rdlink._tcp.l...
531	2.402464	fe80::1ca7:44fe:b6d...	ff02::fb	MDNS	157	Standard query 0x0000 PTR _companion-link._tcp.local, "QM" question PTR _rdlink._tcp.l...
539	2.443392	fe80::5ce1:c1ff:fe...	ff02::fb	MDNS	506	Standard query response 0x0000 PTR, cache flush Android_UL838LD4.local AAAA, cache flu...
540	2.443392	10.10.93.168	224.0.0.251	MDNS	486	Standard query response 0x0000 PTR, cache flush Android_UL838LD4.local AAAA, cache flu...
576	2.653918	10.10.93.104	239.255.255.250	SSDP	167	M-SEARCH * HTTP/1.1
577	2.657748	10.10.93.104	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
578	2.657748	fe80::6837:97ff:fe...	ff02::fb	MDNS	107	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question
623	3.002309	10.10.92.181	10.10.93.255	UDP	90	64740 → 59200 Len=48

Frame 508: 339 bytes on wire (2712 bits), 339 bytes captured (2712 bits) on interface \Device\NPF_{C010246A-3BF0-4D39-AAE3-A940C94EA532}, id 0

Ethernet II, Src: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01), Dst: CloudNetwork_3e:09:e1 (30:03:c8:3e:09:e1)

Destination: CloudNetwork_3e:09:e1 (30:03:c8:3e:09:e1)

Source: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01)

Type: IPv4 (0x0800)

[Stream index: 16]

Internet Protocol Version 4, Src: 10.10.0.26, Dst: 10.10.92.110

User Datagram Protocol, Src Port: 53, Dst Port: 62101

Domain Name System (response)

Packets: 39198 · Displayed: 603 (1.5%) · Dropped: 0 (0.0%) · Profile: Default

Ans – 2)**(a) TCP Connection using Follow [SYN] TCP stream.**

No.	Time	Source	Destination	Protocol	Length	Info
1517	4.554377	10.10.93.234	172.217.24.163	TCP	66	55834 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
1523	4.559420	10.10.93.234	172.217.24.163	TCP	66	50291 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2092	4.694677	10.10.93.234	35.190.80.1	TCP	66	58572 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2448	6.212114	10.10.93.234	52.203.223.25	TCP	66	58574 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2781	6.797827	10.10.93.234	54.192.142.105	TCP	66	53601 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2964	7.148840	10.10.0.26	10.10.92.59	ICMP	106	Destination unreachable (Host administratively prohibited)
3068	7.326930	10.10.93.234	142.251.221.131	TCP	66	62902 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
4835	9.371031	10.10.93.234	52.104.96.55	TCP	66	62903 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
7181	11.768555	10.10.90.246	10.10.92.110	TCP	66	64987 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11309	20.073582	10.10.93.234	10.10.76.216	TCP	66	62904 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
12526	21.081786	10.10.93.234	10.10.76.216	TCP	66	[TCP Retransmission] 62904 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
14323	23.096284	10.10.93.234	10.10.76.216	TCP	66	[TCP Retransmission] 62904 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
14627	24.127703	10.10.93.234	52.203.223.25	TCP	66	62905 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
15717	27.111185	10.10.93.234	10.10.76.216	TCP	66	[TCP Retransmission] 62904 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
15866	27.430339	10.10.93.234	52.182.143.209	TCP	66	62906 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
20379	35.115492	10.10.93.234	10.10.76.216	TCP	66	[TCP Retransmission] 62904 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
25451	42.275618	10.10.90.170	10.10.93.234	TCP	66	64077 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Frame 1517: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{C010246A-3BF0-4D39-AAE3-A940C94EA532}, id 0

Ethernet II, Src: Dell_7a:81:a7 (b4:45:06:7a:81:a7), Dst: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01)

Destination: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01)

Source: Dell_7a:81:a7 (b4:45:06:7a:81:a7)

Type: IPv4 (0x0800)

[Stream index: 17]

Internet Protocol Version 4, Src: 10.10.93.234, Dst: 172.217.24.163

Transmission Control Protocol, Src Port: 55834, Dst Port: 443, Seq: 0, Len: 0

Source Hardware Address (eth.src), 6 bytes

Packets: 39198 · Displayed: 32 (0.1%) · Dropped: 0 (0.0%) · Profile: Default

(b) UDP – dns communication.

No.	Time	Source	Destination	Protocol	Length	Info
3710	8.974668	1.10.10.10	10.10.93.54	DNS	202	Standard query response 0x1085 A www.google.com A 142.251.156.119 A 142.251.150.119 A ...
12545	21.145971	1.10.10.10	10.10.92.50	DNS	166	Standard query response 0x007d HTTPS history.google.com CNAME history.l.google.com SOA...
12553	21.177826	1.10.10.10	10.10.92.50	DNS	212	Standard query response 0xe906 A history.google.com CNAME history.l.google.com A 74.12...
85	0.508459	10.10.0.26	10.10.92.50	DNS	402	Standard query response 0xf0d6 A dns.msftncsi.com A 131.107.255.255 NS ns1-34.azure-dn...
87	0.508350	10.10.0.26	10.10.92.50	DNS	490	Standard query response 0x63c6 A www.msftncsi.com CNAME www.msftncsi.com.edgesuite.net...
299	1.223477	10.10.0.26	10.10.92.168	DNS	398	Standard query response 0x6dc1 A play.googleapis.com A 216.239.36.223 A 216.239.38.223...
505	2.301556	10.10.0.26	10.10.92.110	DNS	339	Standard query response 0x44eb A chat.google.com A 142.250.205.110 NS ns1.google.com NL...
508	2.311084	10.10.0.26	10.10.92.110	DNS	339	Standard query response 0x1d00 A chat.google.com A 142.250.205.110 NS ns2.google.com NL...
723	3.731728	10.10.0.26	10.10.92.59	DNS	380	Standard query response 0x0981 A alt8-mtalk.google.com CNAME alt8.mobile-gtalk4.l.goog...
2136	4.784278	10.10.0.26	10.10.92.97	DNS	412	Standard query response 0x9917 A firebase logging-pa.googleapis.com A 216.239.38.223 A ...
2220	5.128233	10.10.0.26	10.10.93.96	DNS	350	Standard query response 0x6ab2 A app-measurement.com A 142.251.223.238 NS ns1.google.c...
2384	5.902178	10.10.0.26	10.10.93.234	DNS	173	Standard query response 0xc8dd HTTPS applet-bundles.grammarly.net SOA ns-1253.awsdns-2...
2385	5.902178	10.10.0.26	10.10.93.234	DNS	465	Standard query response 0xa69f A applet-bundles.grammarly.net A 54.192.142.105 A 54.19...
2386	5.903205	10.10.0.26	10.10.93.234	DNS	465	Standard query response 0x11ff A applet-bundles.grammarly.net A 54.192.142.70 A 54.192...
2539	6.419446	10.10.0.26	10.10.92.59	DNS	226	Standard query response 0xa65e A www.baidu.com CNAME www.a.shifen.com CNAME www.wshife...
2779	6.796437	10.10.0.26	10.10.93.234	DNS	173	Standard query response 0xa4ea AAAA applet-bundles.grammarly.net SOA ns-1253.awsdns-28...
2780	6.796437	10.10.0.26	10.10.93.234	DNS	173	Standard query response 0x03c6 AAAA applet-bundles.grammarly.net SOA ns-1253.awsdns-28...

> Frame 3710: 202 bytes on wire (1616 bits), 202 bytes captured (1616 bits) on interface \Device\NPF_{C010246A-38F0-4D39-AAE3-A940C94EA532}, id 0
 > Ethernet II, Src: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01), Dst: 0e:ef:88:37:98:2c (0e:ef:88:37:98:2c)
 > Destination: 0e:ef:88:37:98:2c (0e:ef:88:37:98:2c)
 > Source: HewlettPacka_ba:4a:01 (d8:94:03:ba:4a:01)
 Type: IPv4 (0x0800)
 [Stream index: 26]
 > Internet Protocol Version 4, Src: 1.10.10.10, Dst: 10.10.93.54
 > User Datagram Protocol, Src Port: 53, Dst Port: 40084
 > Domain Name System (response)

0000 0e ef 88 37 98
 0010 00 bc 51 58 00
 0020 5d 36 00 35 9c
 0030 00 00 00 00 00
 0040 65 03 63 6f 6d
 0050 00 00 00 d6 00
 0060 00 00 00 d6 00
 0070 00 00 00 d6 00
 0080 00 00 00 d6 00
 0090 00 00 00 d6 00
 00a0 00 00 00 d6 00
 00b0 00 00 00 d6 00
 00c0 00 00 00 d6 00

Source Hardware Address (eth.src), 6 bytes Packets: 39198 · Displayed: 231 (0.6%) · Dropped: 0 (0.0%) Profile: Default

Ans – 3) In TCP there are packets of types [SYN] , [SYN, ACK] and [ACK], this confirms the 3-way handshake.

However, in UDP there is no such organized connection. We have only DNS query and Response.

Part 1: HTTP Sniffing and Security awareness —

Ans – 1) <http://testaspnet.vulnweb.com/login.aspx> has some server side error so I will have to work only based on **GET** and **Internal Server Error**.

(a) HTTP Request packet GET

The screenshot shows a Wireshark packet capture of an HTTP GET request. The packet list on the left shows several packets, with packet 10387 (6.208749s) selected, which is an HTTP 200 OK response. The packet details pane on the right shows the Hypertext Transfer Protocol section expanded, displaying the request line: GET /login.aspx HTTP/1.1. The Host is testaspnet.vulnweb.com. The User-Agent is Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36. The Accept-Encoding is gzip, deflate. The Accept-Language is en-US,en;q=0.9,hi;q=0.8. The full request URI is http://testaspnet.vulnweb.com/login.aspx.

(b) HTTP Response packet 200 OK (text/plain)

The screenshot shows a Wireshark packet capture of an HTTP 200 OK response. The packet list on the left shows several packets, with packet 10387 (6.208749s) selected, which is an HTTP 200 OK response. The packet details pane on the right shows the Hypertext Transfer Protocol section expanded, displaying the response line: HTTP/1.1 200 OK. The Content-Length is 14. The Date is Sun, 12 Apr 2026 14:13:29 GMT. The Content-Type is text/plain. The Cache-Control is max-age=30, must-revalidate. The X-Cache is MISS from localhost. The Connection is close. The File Data is 14 bytes. The Line-based text data is text/plain (1 lines).

Ans – 2)

- (a) Requested URL from HTTP request packet details -
<http://testaspnet.vulnweb.com/login.aspx>
- (b) Server Response code from HTTP Response packets –
500 Internal Server Error and 200 OK.

Part 2: Sniffing based script –

Ans – 1) Filters

- (b) “txtUsername” and “txtPassword” using “HTTP” site.

The image shows a Wireshark packet capture of an HTTP POST request. The packet list pane shows a POST request from 10.10.93.234 to 10.10.93.23. The packet details pane shows the request body with form items: txtUsername = 'goku' and txtPassword = 'son'. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
4776	7.734154	13.225.103.61	10.10.93.23	MPEG	1134	HTTP/1.1 200 OK Program Association Table (PAT) Program Map Table (PMT) AUD_N
4972	7.932672	192.168.2.190	10.10.93.47	HTTP	587	HTTP/1.1 200 OK (text/html)
5588	8.844710	23.155.129.172	10.10.93.234	HTTP	579	HTTP/1.1 200 OK (text/html)
5657	9.145653	146.75.118.172	10.10.93.118	HTTP	107	HTTP/1.1 503 Service Unavailable (text/html)
8180	11.604333	184.85.220.72	10.10.93.23	HTTP	1041	HTTP/1.1 200 OK (application/x-mpegurl)
8295	11.810867	13.225.103.61	10.10.93.23	MPEG	410	HTTP/1.1 200 OK Program Association Table (PAT) Program Map Table (PMT) AUD_N
8658	12.997082	104.18.18.219	10.10.92.39	HTTP	60	HTTP/1.1 200 OK (application/x-mpegurl)
8757	13.223457	103.178.50.6	10.10.92.39	MPEG	1150	HTTP/1.1 200 OK Program Association Table (PAT) Program Map Table (PMT) AUD_N
10968	15.077980	104.18.18.219	10.10.92.39	HTTP	60	HTTP/1.1 200 OK (application/x-mpegurl)
11041	15.253755	103.178.50.6	10.10.92.39	MPEG	1079	HTTP/1.1 200 OK Program Association Table (PAT) Program Map Table (PMT) AUD_N
11061	15.433125	10.10.93.234	23.155.129.172	HTTP	759	POST /login/login_results.asp HTTP/1.1 (application/x-www-form-urlencoded)
11174	15.519441	fe80::5888:716d:eea... fe80::b813:a9b:d63d...	10.10.93.96	HTTP/XML	985	HTTP/1.1 200
11257	15.581812	142.251.222.164	10.10.93.96	HTTP	686	HTTP/1.1 200 OK (PNG)
11266	15.590004	184.85.220.72	10.10.93.23	HTTP	1041	HTTP/1.1 200 OK (application/x-mpegurl)
11269	15.591613	172.217.24.132	10.10.93.96	HTTP	1014	HTTP/1.1 200 OK (JPEG JFIF image)
11277	15.642032	172.217.24.4	10.10.93.96	HTTP	587	HTTP/1.1 200 OK (JPEG JFIF image)
11282	15.647619	142.251.222.164	10.10.93.96	HTTP	364	HTTP/1.1 200 OK (PNG)

Referer: http://vbsca.ca/login/login.asp\r\n
 Accept-Encoding: gzip, deflate\r\n
 Accept-Language: en-US,en;q=0.9,hi;q=0.8\r\n
 Cookie: ASPSESSIONIDQSRASBR=0MPIEAOOGDFKNODMLBIEKAC\r\n
 \r\n
 [Response in frame: 13986]
 [Full request URI: http://vbsca.ca/login/login_results.asp]
 File Data: 32 bytes

HTML Form URL Encoded: application/x-www-form-urlencoded
 > Form item: "txtUsername" = "goku"
 > Form item: "txtPassword" = "son"

Text item (text), 15 bytes

Packets: 31598 · Displayed: 85 (0.3%) · Dropped: 0 (0.0%) Profile: Default

- (c) HTTPS / TLS packets after printing unlike the HTTP items as encoded and appear to be random stream of characters via some sort of encoding (SHA256 most likely).